E2 Solutions: complex numbers

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All results were checked with SymPy.

Practice

1. (a) $3 - 12i + 2i - 8i^2 = 11 - 10i$.

(b) $\dfrac{(2-i)(1-3i)}{(1+3i)(1-3i)} = \dfrac{2 - 6i - i + 3i^2}{10} = -\dfrac{1}{10} - \dfrac{7}{10}i$.

(c) $2031 = 4 \cdot 507 + 3$, so $i^{2031} = i^3 = -i$.

2. $z = \dfrac{4 \pm \sqrt{16 - 52}}{2} = \dfrac{4 \pm 6i}{2} = 2 \pm 3i$. Check: $(2+3i)^2 - 4(2+3i) + 13 = (-5 + 12i) - 8 - 12i + 13 = 0$. ✓

3. (a) $|z| = \sqrt{1+3} = 2$. Second quadrant, $\tan\varphi = -\sqrt{3}$: $\arg z = \frac{2\pi}{3}$.

(b) $|z| = \sqrt{18} = 3\sqrt{2}$. Third quadrant, $\tan\varphi = 1$: $\arg z = -\frac{3\pi}{4}$.

4. $4(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}) = 2\sqrt{3} + 2i$. $2(\cos\frac{3\pi}{4} - i\sin\frac{3\pi}{4}) = -\sqrt{2} - i\sqrt{2}$.

5. (a) Circle with centre $2i$ (the point $(0, 2)$) and radius 3.

(b) Points equidistant from 1 and $-i$: the perpendicular bisector of the segment from $(1, 0)$ to $(0, -1)$. With $z = x + iy$: $(x-1)^2 + y^2 = x^2 + (y+1)^2 \iff -2x = 2y \iff y = -x$.

(c) The open half-line from 0 through $1 + i$ (the origin itself excluded, since $\arg 0$ is undefined).

(d) Points at least as close to 1 as to -1: the closed half-plane $\operatorname{Re} z \geq 0$.

6. $1 + i = \sqrt{2}e^{i\pi/4}$, so $(1+i)^{10} = 2^5 e^{10i\pi/4} = 32e^{5i\pi/2} = 32e^{i\pi/2} = 32i$.

7. $8i = 8e^{i\pi/2}$. $z = 2e^{i(\pi/6 + 2k\pi/3)}$, $k = 0, 1, 2$:

- $2e^{i\pi/6} = \sqrt{3} + i$
- $2e^{5i\pi/6} = -\sqrt{3} + i$
- $2e^{3i\pi/2} = -2i$

They form an equilateral triangle inscribed in the circle of radius 2.

8. Let $z = a + bi$, $w = c + di$.

(a) $z\bar{z} = (a+bi)(a-bi) = a^2 - (bi)^2 = a^2 + b^2 = |z|^2$.

(b) See notes §3: both sides equal $(ac - bd) - (ad + bc)i$.

(c) $z = \bar{z} \iff a + bi = a - bi \iff 2b = 0 \iff z$ is real.

9. $a^2 - b^2 = 5$ and $2ab = -12$. Also $|a + bi|^2 = |5 - 12i| = 13$, so $a^2 + b^2 = 13$. Adding: $a^2 = 9$; subtracting: $b^2 = 4$. Since $ab = -6 < 0$: $(a, b) = (3, -2)$ or $(-3, 2)$.

10. Rotation about c by $90°$: $z \mapsto c + i(z - c)$. Here $z - c = 3$, so the image is $1 + i + 3i = 1 + 4i$.

Exam-level

11. (a) With $c = \cos\varphi$, $s = \sin\varphi$:

$$\cos 3\varphi + i\sin 3\varphi = (c + is)^3 = c^3 + 3c^2(is) + 3c(is)^2 + (is)^3 = (c^3 - 3cs^2) + i(3c^2 s - s^3).$$

So $\cos 3\varphi = c^3 - 3c(1 - c^2) = 4c^3 - 3c$ and $\sin 3\varphi = 3(1 - s^2)s - s^3 = 3s - 4s^3$.

(b) A cubic has at most three roots. With $x = \cos\varphi$ the equation becomes $\cos 3\varphi = \frac{1}{2}$, so $3\varphi = \pm\frac{\pi}{3} + 2k\pi$, $\varphi = \pm\frac{\pi}{9} + \frac{2k\pi}{3}$. This gives three different values of $\cos\varphi$:

$$x = \cos\frac{\pi}{9}, \quad x = \cos\frac{5\pi}{9}, \quad x = \cos\frac{7\pi}{9}.$$

(Numerically 0.940, -0.174, -0.766; all satisfy $4x^3 - 3x = 0.5$.) Since there are three distinct values and a cubic has at most three roots, these are all the solutions.

12. (a) $P(1) = 1 - 3 + 7 - 5 = 0$.

(b) Polynomial division: $P(z) = (z-1)(z^2 - 2z + 5)$. Check: $z^3 - 2z^2 + 5z - z^2 + 2z - 5 = z^3 - 3z^2 + 7z - 5$. ✓ The quadratic gives $z = \frac{2 \pm \sqrt{-16}}{2} = 1 \pm 2i$. Roots: 1, $1 + 2i$, $1 - 2i$.

(c) P has real coefficients, so by the conjugate root theorem $P(\bar{z}_0) = \overline{P(z_0)} = 0$ whenever $P(z_0) = 0$.

13. (a) $-4 = 4e^{i\pi}$, so $z = \sqrt{2}e^{i(\pi/4 + k\pi/2)}$: $1 + i$, $-1 + i$, $-1 - i$, $1 - i$.

(b) Pair conjugates: $(z - (1+i))(z - (1-i)) = z^2 - 2z + 2$ and $(z - (-1+i))(z - (-1-i)) = z^2 + 2z + 2$. So

$$z^4 + 4 = (z^2 - 2z + 2)(z^2 + 2z + 2).$$

(Check: $(z^2+2)^2 - (2z)^2 = z^4 + 4z^2 + 4 - 4z^2$. ✓ This is the Sophie Germain identity.)

14. (a) $\omega \neq 1$ and $\omega^5 = 1$, so the geometric sum equals $\dfrac{\omega^5 - 1}{\omega - 1} = 0$.

(b) Real parts: $1 + \cos\frac{2\pi}{5} + \cos\frac{4\pi}{5} + \cos\frac{6\pi}{5} + \cos\frac{8\pi}{5} = 0$. Since $\cos\frac{8\pi}{5} = \cos\frac{2\pi}{5}$ and $\cos\frac{6\pi}{5} = \cos\frac{4\pi}{5}$ (as $\cos(2\pi - x) = \cos x$): $1 + 2\cos\frac{2\pi}{5} + 2\cos\frac{4\pi}{5} = 0$, which gives the result.

(c) $c + 2c^2 - 1 = -\frac{1}{2}$, so $4c^2 + 2c - 1 = 0$ and $c = \dfrac{-2 \pm \sqrt{20}}{8} = \dfrac{-1 \pm \sqrt{5}}{4}$. Since $0 < \frac{2\pi}{5} < \frac{\pi}{2}$, $c > 0$: $\cos\frac{2\pi}{5} = \dfrac{\sqrt{5} - 1}{4} \approx 0.309$.

15. $p = 6$, $q = -20$. Put $x = u + v$ with $uv = -2$ and $u^3 + v^3 = 20$. Then $u^3 v^3 = -8$, so u^3, v^3 are roots of $t^2 - 20t - 8 = 0$: $t = 10 \pm \sqrt{108} = 10 \pm 6\sqrt{3}$. Since $(1+\sqrt{3})^3 = 1 + 3\sqrt{3} + 9 + 3\sqrt{3} = 10 + 6\sqrt{3}$ and similarly $(1-\sqrt{3})^3 = 10 - 6\sqrt{3}$, take $u = 1 + \sqrt{3}$, $v = 1 - \sqrt{3}$ (check $uv = 1 - 3 = -2$ ✓). So $x = 2$. Other roots: $x^3 + 6x - 20 = (x-2)(x^2 + 2x + 10)$, giving $x = -1 \pm 3i$. (Or from Cardano: $u\omega + v\omega^2$ and $u\omega^2 + v\omega$ with $\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$; e.g. $u\omega + v\omega^2 = -\frac{1}{2}(u+v) + \frac{\sqrt{3}}{2}i(u-v) = -1 + \frac{\sqrt{3}}{2}i \cdot 2\sqrt{3} = -1 + 3i$.)

16. (a) $(2+i)^3 = 8 + 3 \cdot 4 \cdot i + 3 \cdot 2 \cdot i^2 + i^3 = 8 + 12i - 6 - i = 2 + 11i$.

(b) Taking conjugates, $(2-i)^3 = 2 - 11i$. With $u = 2 + i$, $v = 2 - i$ we have $uv = 5 = \frac{p}{3}$ for $p = 15$ (Cardano for $x^3 = px + q$ needs $uv = \frac{p}{3}$), so $x = u + v = 4$. Check: $64 = 60 + 4$. ✓ Then $x^3 - 15x - 4 = (x-4)(x^2 + 4x + 1)$, so the other solutions are $x = -2 \pm \sqrt{3}$, both real. All three roots are real, yet the formula needs non-real numbers: the irreducible case.

17. With $z = x + iy$: $(x-3)^2 + y^2 = 4(x^2 + y^2) \iff 3x^2 + 3y^2 + 6x - 9 = 0 \iff x^2 + 2x + y^2 = 3 \iff (x+1)^2 + y^2 = 4$. A circle with centre -1 and radius 2 (a circle of Apollonius for the points 3 and 0 with ratio 2). Check: $z = 1$ gives $|1 - 3| = 2 = 2|1|$ ✓; $z = -3$ gives $6 = 2 \cdot 3$ ✓.

18. (a) In an equilateral triangle $|w| = |z|$ and the angle from z to w at the vertex 0 is $60°$; anticlockwise means $\arg w = \arg z + \frac{\pi}{3}$. So $w = e^{i\pi/3}z$.

(b) $\rho = e^{i\pi/3}$ satisfies $\rho^3 = e^{i\pi} = -1$, so $0 = \rho^3 + 1 = (\rho+1)(\rho^2 - \rho + 1)$, and $\rho \neq -1$, so $\rho^2 - \rho + 1 = 0$. Then $z^2 + w^2 - zw = z^2(1 + \rho^2 - \rho) = 0$.

19. (a) $c = (1+i)c + 2 \iff -ic = 2 \iff c = \frac{2}{-i} = 2i$.

(b) $f(z) - 2i = (1+i)(z - 2i)$ and $1 + i = \sqrt{2}e^{i\pi/4}$: a rotation by $45°$ about $2i$ followed by scaling by factor $\sqrt{2}$ from $2i$ (a spiral similarity).

(c) Four times: rotation by $180°$ and scaling by 4 about $2i$: $f^4(z) - 2i = (1+i)^4(z - 2i) = -4(z - 2i)$. For $z = 0$: $f^4(0) = 2i + 8i = 10i$. Check: $f(0) = 2$, $f(2) = 4 + 2i$, $f(4 + 2i) = (4 + 2i + 4i - 2) + 2 = 4 + 6i$, $f(4 + 6i) = (4 + 6i + 4i - 6) + 2 = 10i$. ✓

20. For real x: $e^{(1+i)x} = e^x(\cos x + i \sin x)$, whose real part is $e^x \cos x$. An antiderivative of $e^{(1+i)x}$ is $\dfrac{e^{(1+i)x}}{1+i} = \dfrac{1-i}{2}e^x(\cos x + i \sin x)$. Its real part: $\frac{1}{2}e^x(\cos x + \sin x)$. So

$$\int e^x \cos x \, dx = \tfrac{1}{2}e^x(\cos x + \sin x) + C.$$

Check: $\frac{d}{dx}\left[\frac{1}{2}e^x(\cos x + \sin x)\right] = \frac{1}{2}e^x(\cos x + \sin x) + \frac{1}{2}e^x(-\sin x + \cos x) = e^x \cos x$. ✓ (Taking real parts commutes with integrating because integrating a complex function means integrating real and imaginary parts separately.)

Challenge

21. (a) $z^n - 1 = (z-1)(z^{n-1} + \cdots + 1)$, and $z^n - 1$ has the n distinct roots $1, \omega, \ldots, \omega^{n-1}$, so $z^n - 1 = (z-1)(z-\omega)\cdots(z-\omega^{n-1})$. Cancel the factor $z - 1$ (as polynomials).

(b) Substitute $z = 1$: left side $1 + 1 + \cdots + 1 = n$.

(c) $1 - e^{i\theta} = e^{i\theta/2}(e^{-i\theta/2} - e^{i\theta/2}) = e^{i\theta/2} \cdot (-2i \sin\frac{\theta}{2})$, so $|1 - e^{i\theta}| = 2|\sin\frac{\theta}{2}| = 2\sin\frac{\theta}{2}$ for $0 < \theta < 2\pi$. Taking moduli in (b) with $\theta = \frac{2k\pi}{n}$: $\prod_{k=1}^{n-1} 2\sin\frac{k\pi}{n} = n$. (Numerical check for $n = 7$: product $= 7.000$.)

22. Expand, using $\omega^3 = 1$ and $1 + \omega + \omega^2 = 0$:

$$(a+\omega b+\omega^2 c)(a+\omega^2 b+\omega c) = a^2+b^2+c^2+(\omega+\omega^2)(ab+bc+ca) = a^2+b^2+c^2-ab-bc-ca.$$

(The ab terms: $\omega^2 ab + \omega ab$; bc: $\omega \cdot \omega\, bc + \omega^2\omega^2\, bc = (\omega^2 + \omega^4)bc = (\omega^2 + \omega)bc$; ca: $\omega ca + \omega^2 ca$.) So the condition holds iff $a + \omega b + \omega^2 c = 0$ or $a + \omega^2 b + \omega c = 0$. Now $a + \omega b + \omega^2 c = 0$ with $\omega^2 = -1 - \omega$ becomes $a - c = \omega(c - b)$, i.e. $a - c = -\omega(b - c)$, and $-\omega = e^{i\pi}e^{2i\pi/3} = e^{-i\pi/3}$. So $a - c$ is $b - c$ rotated by $-60°$ about c: then $|a - c| = |b - c|$ and the angle at c is $60°$, so the triangle is equilateral. (Example: $c = 0$, $b = 1$, $a = e^{-i\pi/3}$; then a, b, c is traversed anticlockwise.) The second factor gives, in the same way, $a - c = e^{i\pi/3}(b - c)$: the equilateral triangles with the opposite orientation. Conversely every equilateral triangle is clockwise or anticlockwise, so it makes one factor zero. (Degenerate case $a = b = c$ satisfies both.)

23. Let $S = \sum_{k=1}^{n} e^{ik\theta}$. Since $e^{i\theta} \neq 1$:

$$S = e^{i\theta}\frac{e^{in\theta} - 1}{e^{i\theta} - 1} = e^{i\theta}\frac{e^{in\theta/2} \cdot 2i\sin\frac{n\theta}{2}}{e^{i\theta/2} \cdot 2i\sin\frac{\theta}{2}} = e^{i(n+1)\theta/2}\,\frac{\sin\frac{n\theta}{2}}{\sin\frac{\theta}{2}}.$$

The real part is $\dfrac{\sin\frac{n\theta}{2}\cos\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}}$, and the real part of S is the cosine sum. (Numerical check $n=5$, $\theta=0.7$: both sides -1.4487.) The imaginary part gives the sine sum for free.

24. If $(z+1)^5=z^5$, take moduli: $|z+1|^5=|z|^5$, so $|z-(-1)|=|z-0|$: z is equidistant from -1 and 0, so it is on the perpendicular bisector $\operatorname{Re} z=-\frac{1}{2}$. Exact solutions: $z=0$ is not a solution, so we may write $\left(\frac{z+1}{z}\right)^5=1$, i.e. $1+\frac{1}{z}=\omega^k$ with $\omega=e^{2\pi i/5}$, $k=1,2,3,4$ ($k=0$ would need $\frac{1}{z}=0$). So $z=\dfrac{1}{\omega^k-1}$. With $\omega^k-1=e^{i\theta/2}\cdot 2i\sin\frac{\theta}{2}$, $\theta=\frac{2k\pi}{5}$:

$$z=\frac{e^{-i\theta/2}}{2i\sin\frac{\theta}{2}}=\frac{-i\cos\frac{\theta}{2}-\sin\frac{\theta}{2}}{2\sin\frac{\theta}{2}}=-\frac{1}{2}-\frac{i}{2}\cot\frac{k\pi}{5},\quad k=1,2,3,4.$$

Four solutions (the equation has degree 4 since the z^5 terms cancel), all with real part $-\frac{1}{2}$. ✓

25. (a) For a positive real base, $a^b=e^{b\ln a}$. Copy this: $i=e^{i\pi/2}$, so a "logarithm" of i is $i\frac{\pi}{2}$, and $i^i=e^{i\cdot i\pi/2}=e^{-\pi/2}\approx 0.2079$.

(b) Also $i=e^{i(\pi/2+2k\pi)}$ for every integer k, so every $i(\frac{\pi}{2}+2k\pi)$ is "a logarithm" of i, giving $i^i=e^{-\pi/2-2k\pi}$. Infinitely many values, all real and positive. Choosing the principal argument ($k=0$) gives the principal value $e^{-\pi/2}$.